Stan changes: code and justification

The revised code does less repeated work and samples the spatial, age and income effects in different coordinates. The examples below show the changes and why they preserve the model. Code excerpts omit unchanged declarations.

1. Select fixed data once

Before: select the same rows and columns at every model evaluation.

```
Z[yes_area, Z_id[j, 1:P[j]]]
  * eta_s[1:P[j], j-1]
```

After: store the selection in transformed data, then reuse it.

```
// transformed data: once per chain
Z_yes_cache[j, :, 1:P[j]] = Z[yes_area, Z_id[j, 1:P[j]]];

// the same term in the linear predictor
Z_yes_cache[j, :, 1:P[j]] * eta_s[1:P[j], j-1]
```

Result: the matrix product is identical. Only the repeated selection is removed. The same change is applied to `X` and the respondent-level IDs.

2. Keep the age and income priors; change how they are sampled

Before: these statements jointly define the age prior for outcome j.

```
sum(xi_age[, j]) ~ normal(0, 0.01 * age_N);
for (i in 2:age_N)
  xi_age[i, j] ~ normal(xi_age[i-1, j], 1);
```

For one column, write `a = xi_age[, j]` and `n = age_N`. The prior penalizes both adjacent differences and the total:

$$p(a) \propto \exp\{ -\tfrac{1}{2} [\textstyle\sum_{i=2}^{n}(a_i - a_{i-1})^2 + (\sum_{i=1}^{n} a_i)^2 / (0.01n)^2] \}$$

If D computes adjacent differences and **1** is a vector of ones, the same penalty is $a^T Q a$, where

$$Q = D^T D + \mathbf{1}\,\mathbf{1}^T / (0.01n)^2.$$

After: build Q_age once, then sample standard-normal variables.

```
// transformed data
age_transform = mdivide_left_tri_low(cholesky_decompose(Q_age),
                  diag_matrix(rep_vector(1.0, age_N)))';
// transformed parameters
matrix[age_N, J-1] xi_age = age_transform * xi_age_raw;
// model
to_vector(xi_age_raw) ~ std_normal();
```

The code sets $Q = LL^T$ and $a = L^{-T}z$, with $z \sim N(0, I)$. Hence

$$a^T Q a = z^T z, \quad \mathrm{Cov}(a) = L^{-T}L^{-1} = Q^{-1}.$$

Result: the reconstructed age effects have exactly the original joint prior. Income uses the same construction. The subsequent multiplication by `xi_age_sd` or `xi_inc_sd` stays as before.

The spatial prior and a worked example

3. Apply the same construction to the spatial effects

For one outcome, let ψ contain the spatial effects. The original prior combines differences along graph edges E with a soft sum constraint for each connected group g, containing n_g areas:

$$p(\psi) \propto \exp\{ -\tfrac{1}{2} [\sum_{(a,b)\in E}(\psi_a - \psi_b)^2 + \sum_g(\sum_{a\in g}\psi_a)^2 / (0.001n_g)^2] \}.$$

The new `Q_psi` contains exactly these two terms. Each edge contributes four entries, because $(\psi_a - \psi_b)^2 = \psi_a^2 + \psi_b^2 - 2\psi_a\psi_b$:

```
Q_psi[node1[e], node1[e]] += 1;
Q_psi[node2[e], node2[e]] += 1;
Q_psi[node1[e], node2[e]] -= 1;
Q_psi[node2[e], node1[e]] -= 1;
```

The sum constraint adds $1 / (0.001n_g)^2$ to every entry in that group's block. In matrix form, with D_E computing edge differences and u_g indicating group membership:

$$Q_\psi = D_E^T D_E + \sum_g u_g u_g^T / (0.001n_g)^2.$$

```
// transformed parameters; psi_transform = inverse(L')
matrix[area_N, J-1] psi = psi_transform * psi_raw;
// model, for each non-reference outcome j
psi_raw[, j] ~ std_normal();
```

Result: this gives the same spatial prior by the identity on page 1. The group sums remain close to zero rather than being fixed at zero. A group with one area still gives its ψ effect a normal prior with standard deviation 0.001. The old prior statements are replaced, since applying them again would count the prior twice.

Why the change helps: two age levels

With two levels, the original prior allows the difference $d = a_2 - a_1$ to vary with standard deviation 1, but the sum $s = a_1 + a_2$ has standard deviation 0.02. These very different scales make the original coordinates difficult for Stan to explore.

Take independent $z_s, z_d \sim N(0, 1)$, and define

$$s = 0.02z_s, \quad d = z_d, \quad a_1 = (s - d)/2, \quad a_2 = (s + d)/2.$$

For $z_s = z_d = 1$, the effects are (-0.49, 0.51). Their sum is 0.02 and their difference is 1. The original quadratic term is $(0.02/0.02)^2 + 1^2 = 2$, exactly the new term $z_s^2 + z_d^2 = 2$. Both therefore give a log-prior contribution of -1, apart from constants.

This illustration uses sum and difference coordinates. The code uses Cholesky coordinates; both reproduce the same joint prior. The new variables are independent under the prior, but the likelihood can still make them dependent in the posterior.

Shared function file. The model still includes `functions.stan` and calls `convolve_bym2()` to combine the spatial effects. `lambda_sd` still enters that function. In the original prior used here, it did not enter the spatial quadratic penalty, so `Q_psi` can be computed from the graph alone.